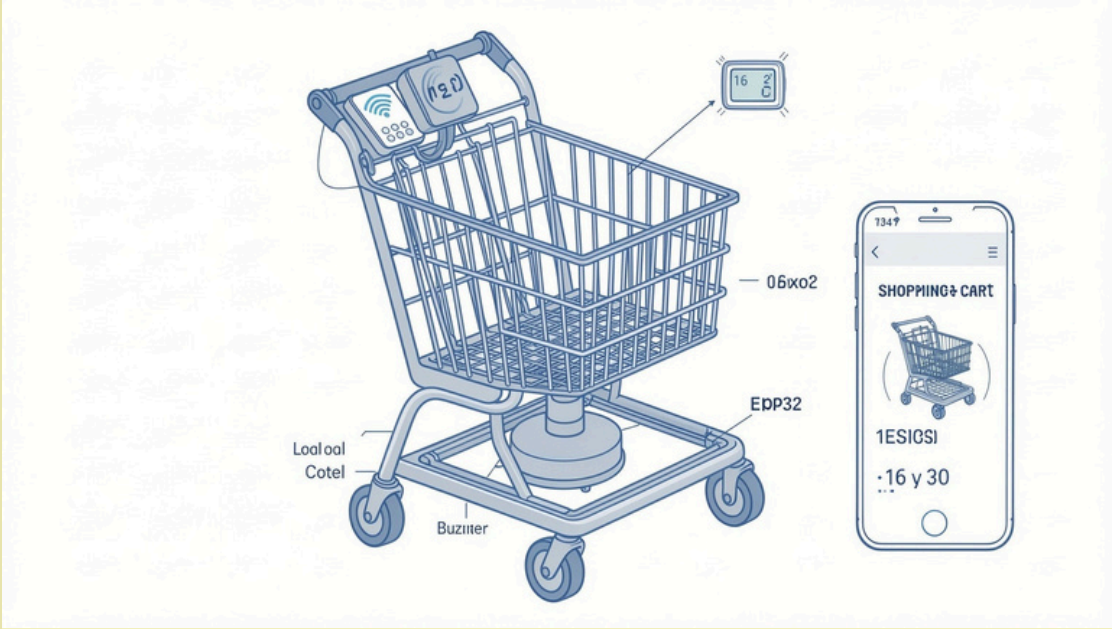
RFID and Weight Sensor-Based Smart Billing Trolley



Team - 2
BODDU KUNAL - 240003020
HARSHITH PASUPULETI - 240003034
DANTULURI SAI HARSHIT VARMA - 240003024
KATHI TAGORE CHANDRA - 240003039
NANAVATH VISHWANATH RATHOD - 240003049

THE PROBLEM

Problem Statement

Traditionalshopping systems requirecustomers to wait at billing counters, where products are scanned and bills are generated individually. This can lead to long queues, delays, and errors during billing.

Project aim

Our project aims to develop a Smart Billing Trolley that simplifies the shopping and billing process using RFID technology and weight sensors.

Problem areas

Slow Billing Process: Manual checkout can increase waiting time, especially during busy hours.
Billing Errors: Missed scans or incorrect product entries can cause discrepancies in the final bill.
Lack of Real-Time Updates: Customers may not have an easy way to view their running bill while shopping .

Our Approach

Integrate RFID-based product identification, weight verification, and automatic bill calculation using an ESP32, with real-time billing information displayed on an LCD and a mobile webpage.

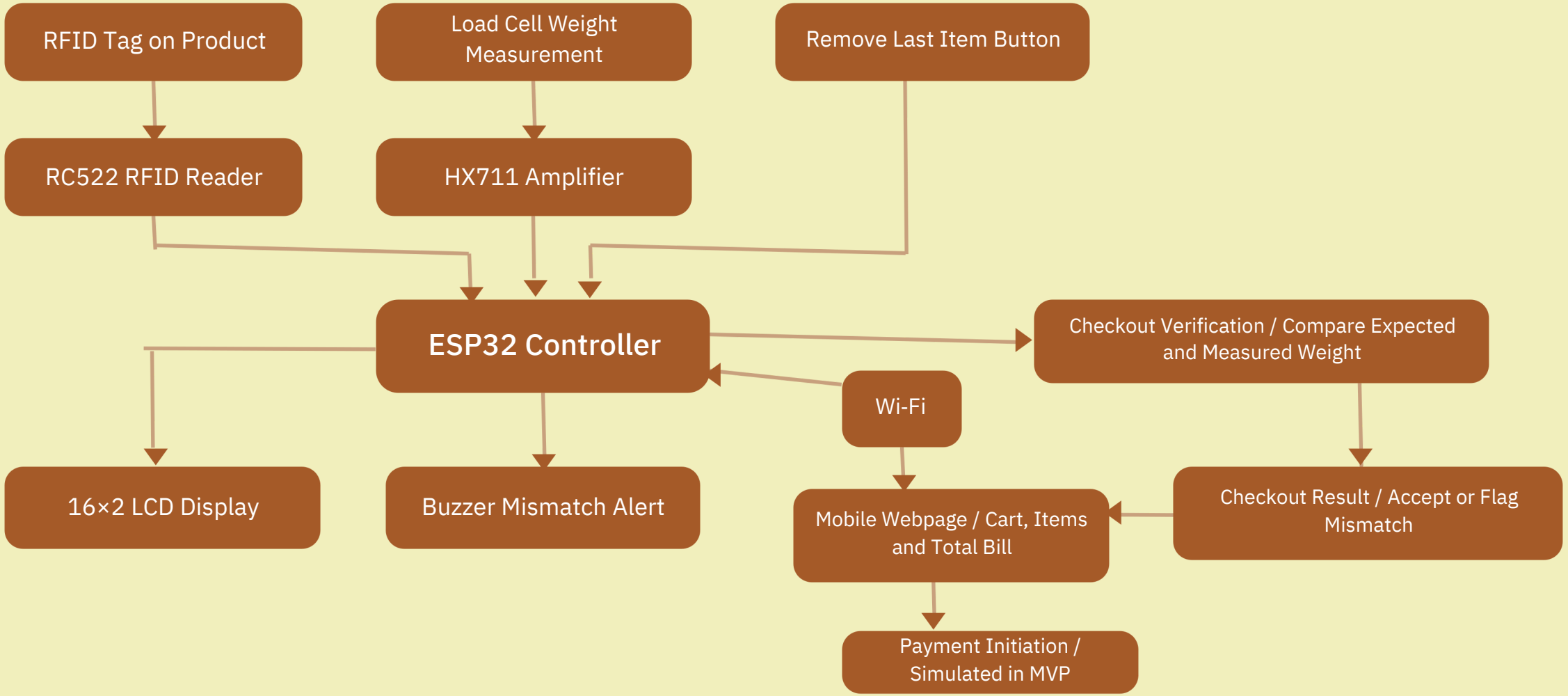
3-STEP FLOW

1 **RFID identifies the product**

2 **Weight sensor verifies the item**

3 **ESP32 updates the bill on the LCD and mobile webpage**

BLOCK DIAGRAM



BUDGET

Component budget

Component	Purpose	Estimated cost
ESP32 development board	Main controller and Wi-Fi	₹373
RC522 RFID reader	Reads RFID tags	₹99
RFID cards/tags	Product identification	₹40
Load cell	Measures weight	₹150
HX711 module	Interfaces load cell with ESP32	₹40
16×2 LCD	Displays product and bill details	₹149
I2C adapter for LCD	Simplifies LCD connections	₹47
Push button	Removes the last scanned item	₹10
Buzzer	Alert on mismatch	₹29
LEDs and resistors	Status indication	₹34
Estimated total		**₹971**

Deliverables

1. RFID-Based Product Identification

Identify products and retrieve their names, prices, and expected weights.

2. Weight Measurement and Verification

Measure product weight using a load cell and HX711 module and flag discrepancies.

3. Automatic Billing System

Calculatethebillandupdateproductquantitiesandthetotal amount.

4. LCD and Buzzer Feedback

Displayproductandbillinginformationandalertuserswhen a mismatch is detected.

5. Wi-Fi-Based Mobile Billing Interface

Display cart items and the total bill on a mobile webpage hosted by the ESP32.

6. Checkout Verification and Payment Demo

Compare expectedandmeasuredtotalweightatcheckoutanddemonstratesimulated payment confirmation.